

LOAN PERFORMANCE & CREDIT RISK ANALYSIS

Using Data Analytics and Machine Learning

An End-to-End Data Analytics & Business Intelligence Project

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Excel

MySQL

Python

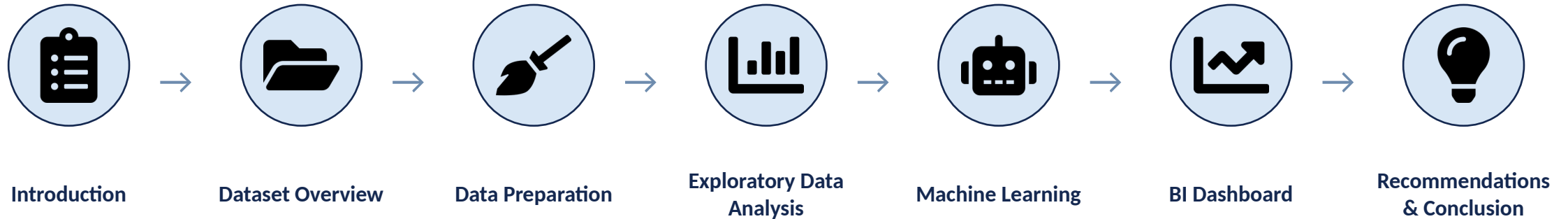
Power BI

Scikit-learn

GitHub

July 2026

Agenda



Sections Covered

- Business Problem & Project Objectives
- Dataset Description & Project Workflow
- Excel Cleaning & SQL Exploratory Analysis
- Python Analysis & Key Statistical Findings
- Model Development, Evaluation & Comparison, Feature Importance

Sections Covered

- Power BI Dashboard Walkthrough & Portfolio Insights
- Key Findings & Strategic Recommendations
- Conclusion & Future Improvements

Business Problem

The Challenge

Financial institutions face increasing pressure to expand lending while minimizing the risk of borrower default. Poor credit risk assessment can lead to financial losses, reduced portfolio quality, and inefficient lending decisions.

Key Challenges

- Difficulty identifying high-risk borrowers before loan approval
- Highly imbalanced loan portfolios where default cases are rare but financially significant
- Traditional assessment methods may overlook important borrower and credit characteristics
- Need for data-driven insights to improve lending decisions and portfolio monitoring



Objective: Use historical loan data to improve credit risk assessment, identify high-risk borrowers, and support data-driven lending decisions.

Project Objectives

Primary Objective

Perform a comprehensive analysis of loan performance and credit risk by integrating data cleaning, exploratory data analysis, business intelligence, and machine learning techniques to support data-driven lending decisions.

1. Raw Data
2. Analysis
3. Insights
4. Predictions
5. Business Decisions

Specific Objectives

- Clean and preprocess the raw loan dataset to improve data quality and consistency
- Perform EDA to understand borrower characteristics, loan performance, and credit behaviour
- Analyze key factors: income, debt-to-income ratio, loan purpose, credit history, delinquency
- Evaluate portfolio health through exposure, default rates, and risk analysis
- Develop interactive Power BI dashboards for executive decision-making
- Build and compare multiple ML models for predicting loan performance
- Identify the most influential factors via feature importance analysis
- Provide actionable business recommendations for credit risk management

Outcome: Build a reliable, data-driven framework for evaluating loan performance, predicting credit risk, and supporting strategic lending decisions.

Dataset Overview

Attribute	Description
Dataset Type	Historical Loan Performance Dataset
Total Records	10,000
Total Features	59 Variables
Target Variable	Loan Status (Performing / Non-Performing)
Analysis Tools	Excel, MySQL, Python, Power BI

**Borrower Information**

**Loan Information**

**Credit Profile**

**Loan Performance**

**Borrower Information**
Annual Income, Homeownership, Employment Length, Verified Income

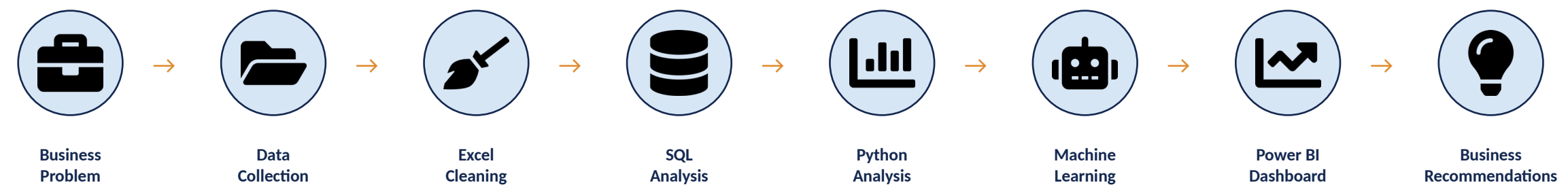
**Loan Information**
Loan Amount, Interest Rate, Loan Purpose, Grade, Term

**Credit Profile**
DTI Ratio, Credit Utilization, Inquiries, Delinquencies, Public Records

**Loan Status: Performing / Non-Performing**

The dataset combines borrower demographics, financial characteristics, loan details, and credit history to support comprehensive credit risk analysis and predictive modeling.

Project Methodology



Phase	Objective
Business Understanding	Define the lending problem and project objectives
Data Preparation	Clean, preprocess, and validate the dataset
Exploratory Analysis	Identify patterns, trends, and borrower characteristics
Predictive Modeling	Develop and evaluate machine learning models
Business Intelligence	Build interactive Power BI dashboards
Decision Support	Generate actionable business insights and recommendations

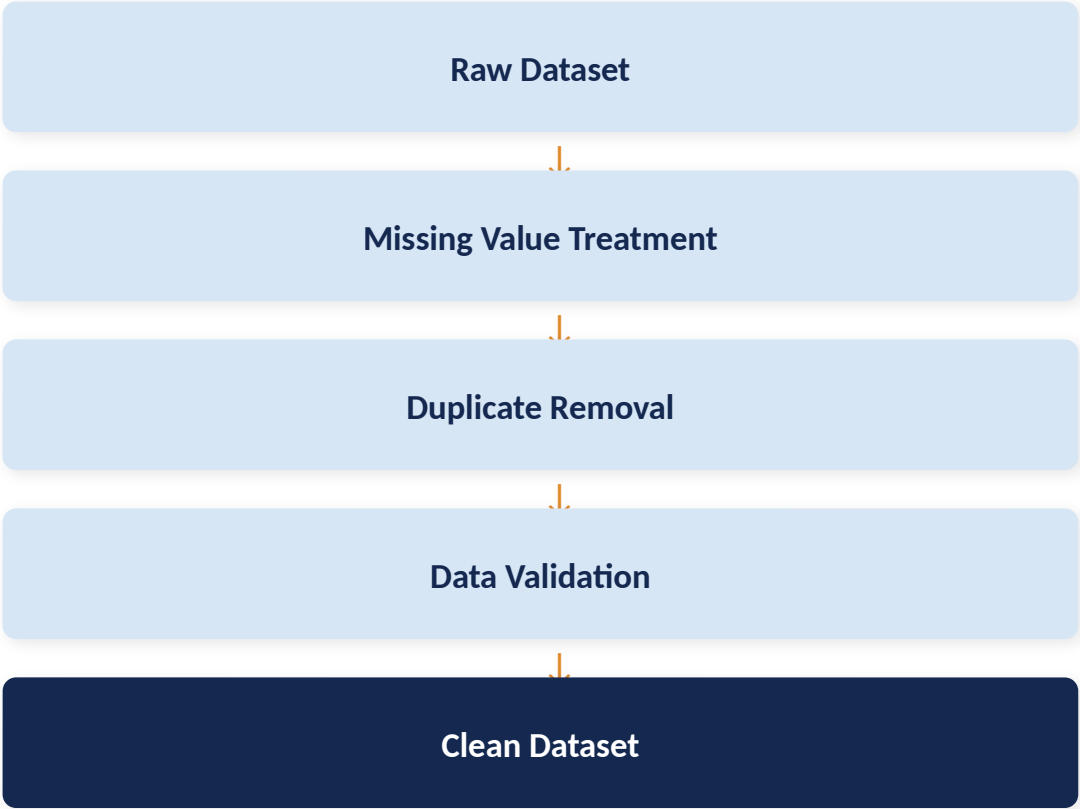
A structured analytics workflow ensured that each stage — from data preparation to predictive modeling and business reporting — contributed to reliable, actionable insights.

Excel Data Cleaning & Preprocessing

Objective

Prepare the raw loan dataset by improving data quality, handling missing values, and ensuring consistency before analysis and predictive modeling.

Task	Action Taken
Missing Values	Treated missing values in employment info and delinquency history
Duplicate Records	Removed duplicate observations to maintain data integrity
Placeholder Values	Replaced missing delinquency values with 999 (No Prior Event)
Numeric Replacement	Replaced missing num_accounts_120d_past_due with 0
Data Validation	Reviewed data types and checked for inconsistencies
Final Dataset	Generated a clean dataset ready for SQL, Python, Power BI



High-quality data is the foundation of reliable analysis. Proper data cleaning improves model accuracy, strengthens business insights, and enhances decision-making.

SQL Exploratory Data Analysis

Business Questions Addressed

- What is the overall composition of the loan portfolio?
- Which loan purposes account for the largest share of lending?
- How are loans distributed across borrower income groups?
- Which loan grades carry the highest credit risk?
- What is the relationship between interest rates and loan performance?

Loan Status	Loans	Exposure
Current	9,375	\$153.4M
Fully Paid	447	\$7.0M
In Grace Period	67	\$1.2M
Late (31-120d)	66	\$1.2M
Late (16-30d)	38	\$0.6M
Charged Off	7	\$0.09M

Income Group	Borrowers
Low Income	2,245
Lower Middle	4,689
Upper Middle	2,610
High Income	456

Loan Purpose	Loans	Funding
Debt Consolidation	5,144	\$90.8M
Credit Card	2,249	\$35.0M
Other	914	\$11.8M
Home Improvement	680	\$11.7M

SQL → Insight Workflow

Clean Dataset → SQL Queries → Business Analysis → Insights → Decision Support

SQL transformed raw loan records into meaningful business insights, providing the analytical foundation for predictive modeling and dashboard reporting.

Key SQL Insights



Portfolio Composition

Portfolio is predominantly performing loans — a small, financially significant non-performing share drives an imbalanced dataset.



Borrower Income Analysis

Most borrowers fall in middle-income categories; higher income did not necessarily mean lower credit risk.



Loan Characteristics

Debt consolidation is the most common purpose; loans concentrate in Grades B and C (moderate risk).



Credit Risk Indicators

Higher DTI, prior delinquencies, and higher interest rates were linked with greater credit risk.



Portfolio Health

The portfolio was generally healthy, though the small share of high-risk loans remains critical to monitor.

SQL Analysis



Portfolio Insights



Borrower Behaviour



Credit Risk Factors



Machine Learning

SQL analysis transformed the cleaned data into actionable business insights and identified the variables most likely to influence loan performance, laying the foundation for the predictive models.

Python Exploratory Data Analysis (EDA)

Objective

Perform statistical analysis and data visualization to understand borrower characteristics, loan performance, and relationships between key financial variables.

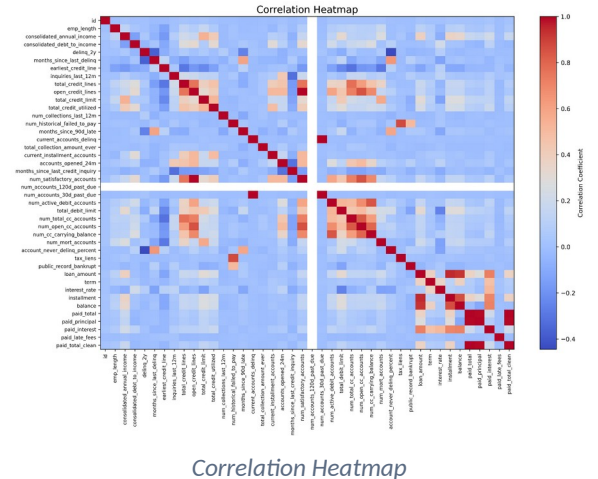
Distribution Analysis

Loan Amount, Income, Interest Rate, DTI

Relationship Analysis

Income vs Loan Amount, Rate vs Status

Correlation Analysis



Annual income showed only a moderate relationship with loan amount. Interest rate and DTI exhibited stronger relationships with loan performance. The target was highly imbalanced — performing loans ≈98% of the portfolio — which drove the choice of the Balanced Logistic Regression model.

Machine Learning Overview

Objective

Develop predictive models to classify loans as Performing or Non-Performing and identify the model offering the most reliable support for credit risk assessment.

- 1. Clean Dataset
- 2. Feature Selection
- 3. Data Encoding
- 4. Train-Test Split
- 5. Model Training
- 6. Model Evaluation
- 7. Model Comparison
- 8. **Best Model Selection**

Model	Purpose
Linear Regression	Baseline regression analysis
Standard Logistic Regression	Initial classification model
Balanced Logistic Regression	Address class imbalance
Standard Random Forest	Tree-based classification
Balanced Random Forest	Feature importance & comparison

Why Multiple Models?

Multiple models were developed and compared to determine which provided the most reliable predictions for loan performance while accounting for the highly imbalanced dataset.

Model evaluation focused on balancing overall predictive performance with the ability to detect non-performing loans, which are critical for effective credit risk management.

Linear Regression Analysis

Why Linear Regression?

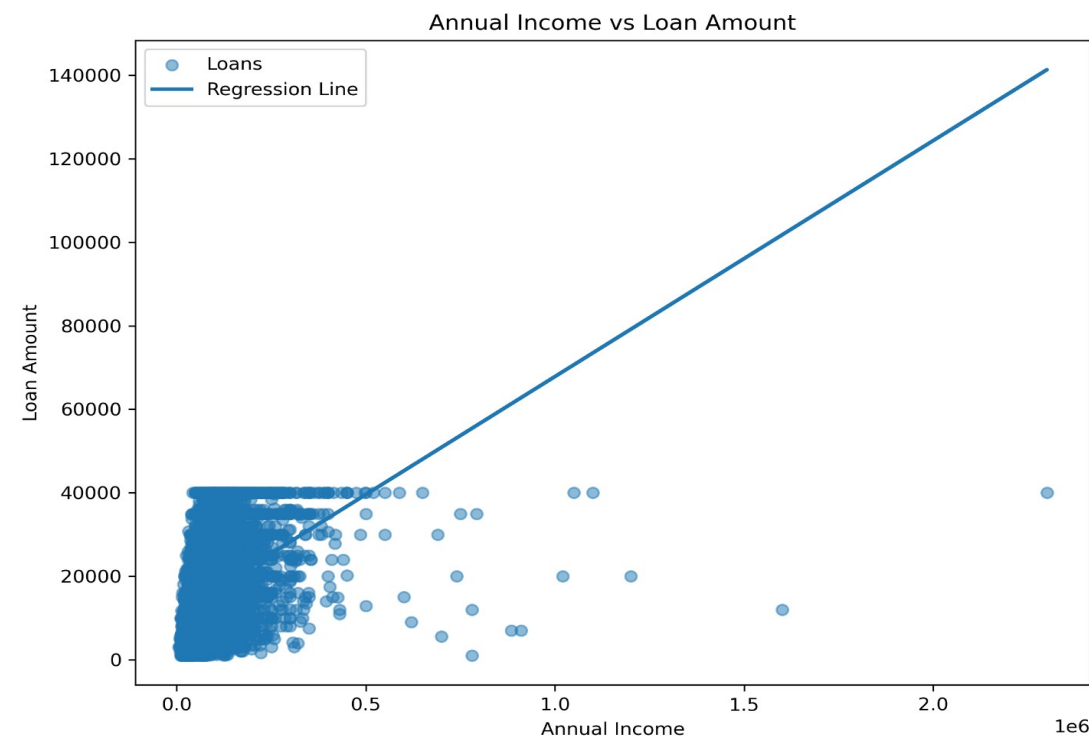
- Explore relationships between financial variables
- Measure the influence of borrower characteristics on loan amount
- Establish a baseline model for comparison

Metric	Result
Correlation (r)	0.381
R ² Score	0.145
P-value	< 0.001

Why It Was Not Selected

- ✗ Predicts continuous values, not categories
- ✗ Cannot classify Performing/Non-Performing
- ✓ Useful baseline for variable relationships

Linear Regression provided valuable baseline insights into borrower behaviour but was not suitable for predicting loan performance because credit risk is fundamentally a classification problem.



Scatter plot: moderate positive relationship between annual income and loan amount

Standard Logistic Regression

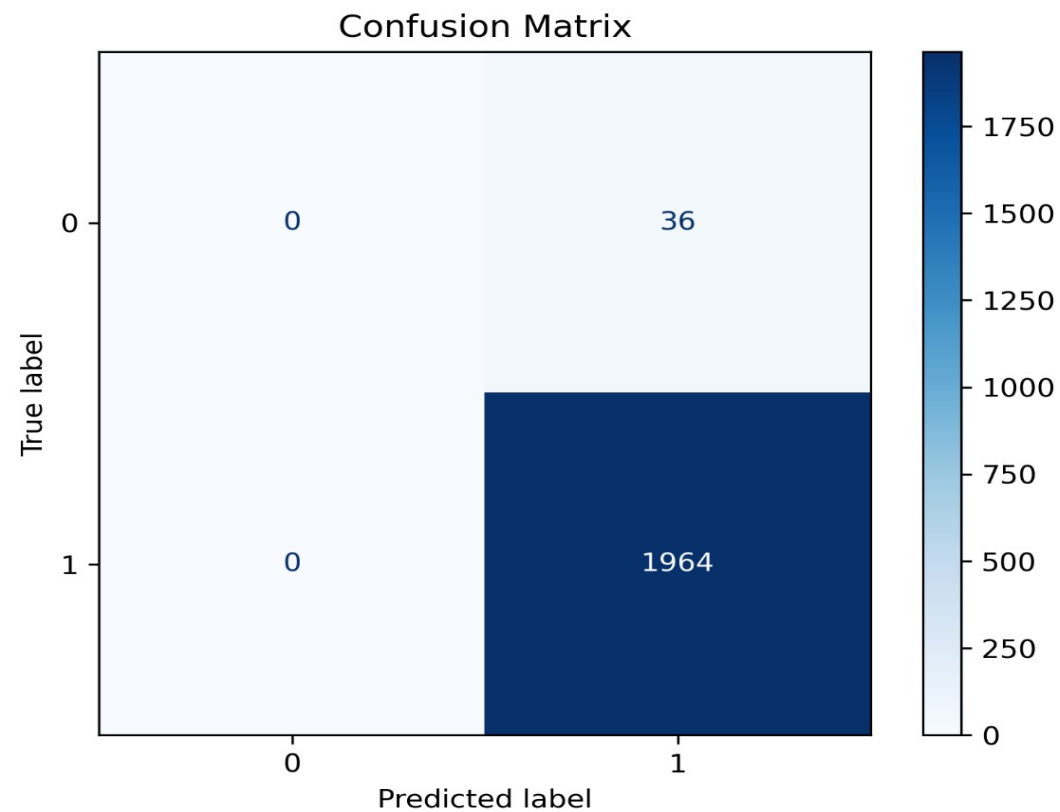
Objective

Develop an initial classification model to predict whether a loan would be Performing or Non-Performing.

Metric	Result
Accuracy	98.2%
Precision	98.2%
Recall	100%
F1 Score	99.1%
ROC-AUC	0.752

36 Default Loans → Detected: 0 → Missed: 36

High accuracy alone does not guarantee a good model. In highly imbalanced datasets, confusion matrices and class-specific performance provide a more reliable assessment than accuracy.



Standard Logistic Regression — Confusion Matrix

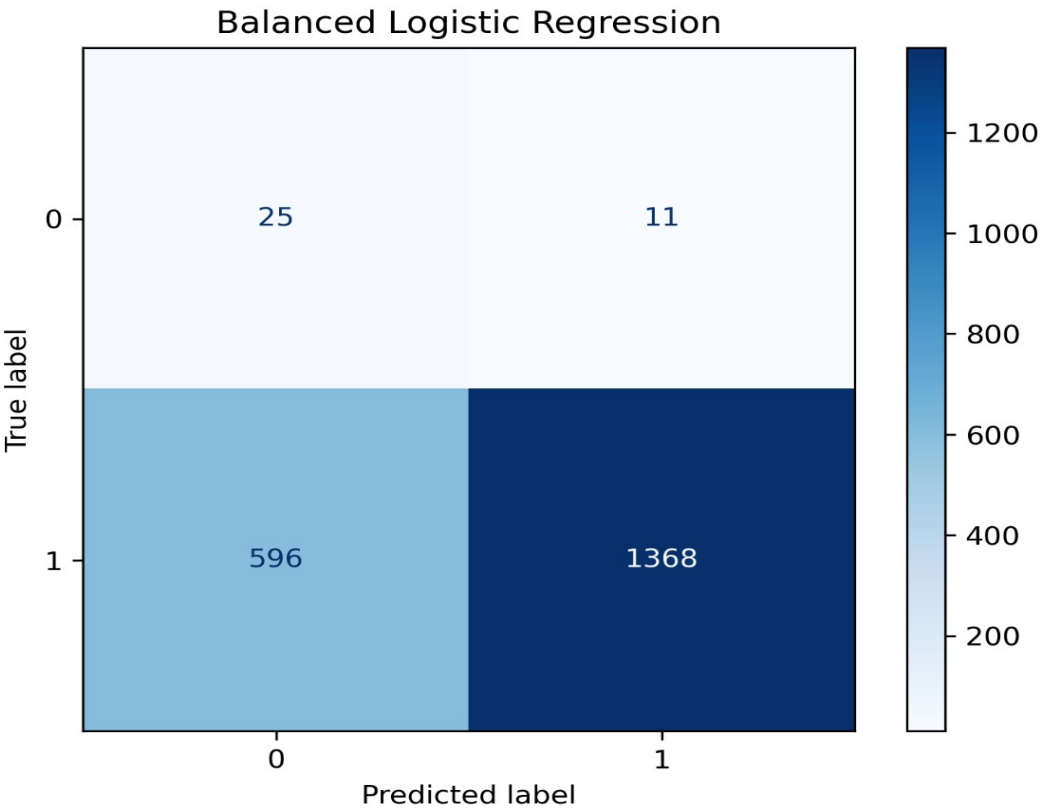
Balanced Logistic Regression (Final Model)

Why a Balanced Model?

The dataset was ≈98.2% Performing vs 1.8% Non-Performing. This severe imbalance biased the standard model toward the majority class. Class weighting was used to improve detection of Non-Performing loans.

Metric	Result
Accuracy	69.7%
Precision	99.2%
Recall	69.7%
F1 Score	81.8%
ROC-AUC	0.762

✓ 25 / 36 Non-Performing Loans Detected (≈69%)



Balanced Logistic Regression — Confusion Matrix

Balanced Logistic Regression was selected because it prioritized identifying high-risk borrowers, making it the most practical model for real-world credit risk assessment despite lower overall accuracy.

Random Forest Model Evaluation

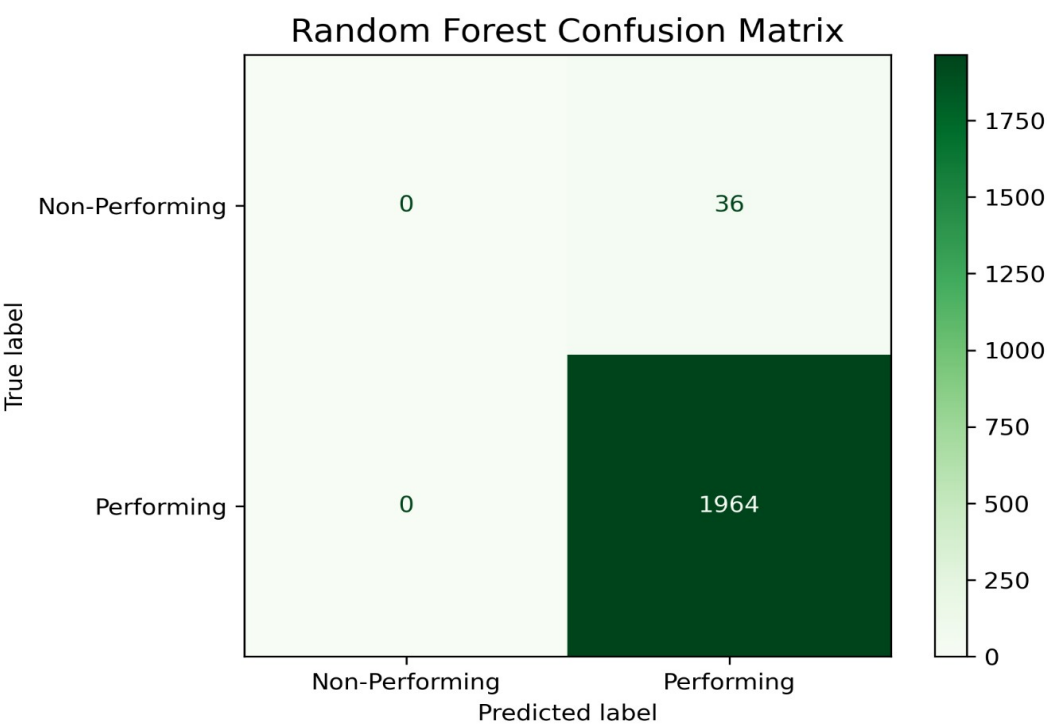
Model	Accuracy	ROC-AUC	NP Detected
Standard Random Forest	98.2%	0.681	0 / 36
Balanced Random Forest	98.2%	0.689	0 / 36

● Strengths

- High Accuracy
- Robust Feature Importance
- Handles Complex Relationships

● Limitations

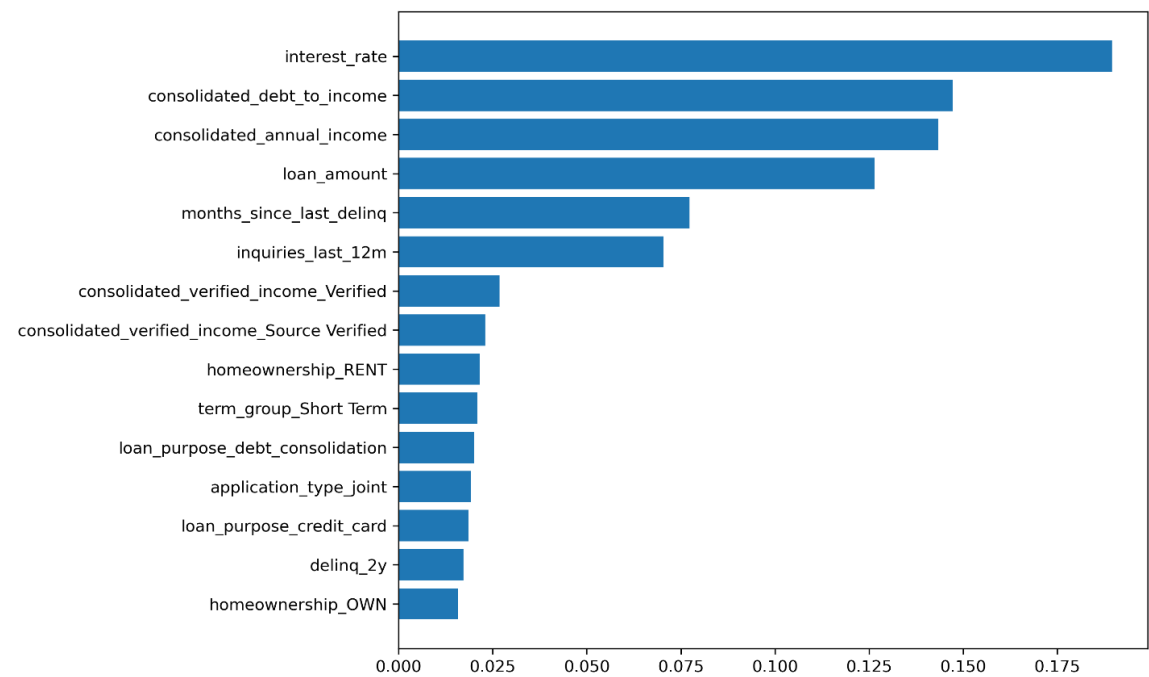
- Detected 0 Non-Performing Loans
- Biased toward Majority Class
- Lower ROC-AUC than Balanced LR



Standard Random Forest — Confusion Matrix

Random Forest demonstrated strong predictive accuracy but failed to identify high-risk borrowers. Its greatest contribution was identifying the most important factors influencing loan performance.

Feature Importance Analysis



Top 10 Feature Importance

Interest Rate

Strongest indicator of borrower risk.

Debt-to-Income Ratio

Measures repayment capacity.

Loan Amount

Represents financial exposure.

Annual Income

Indicates repayment ability.

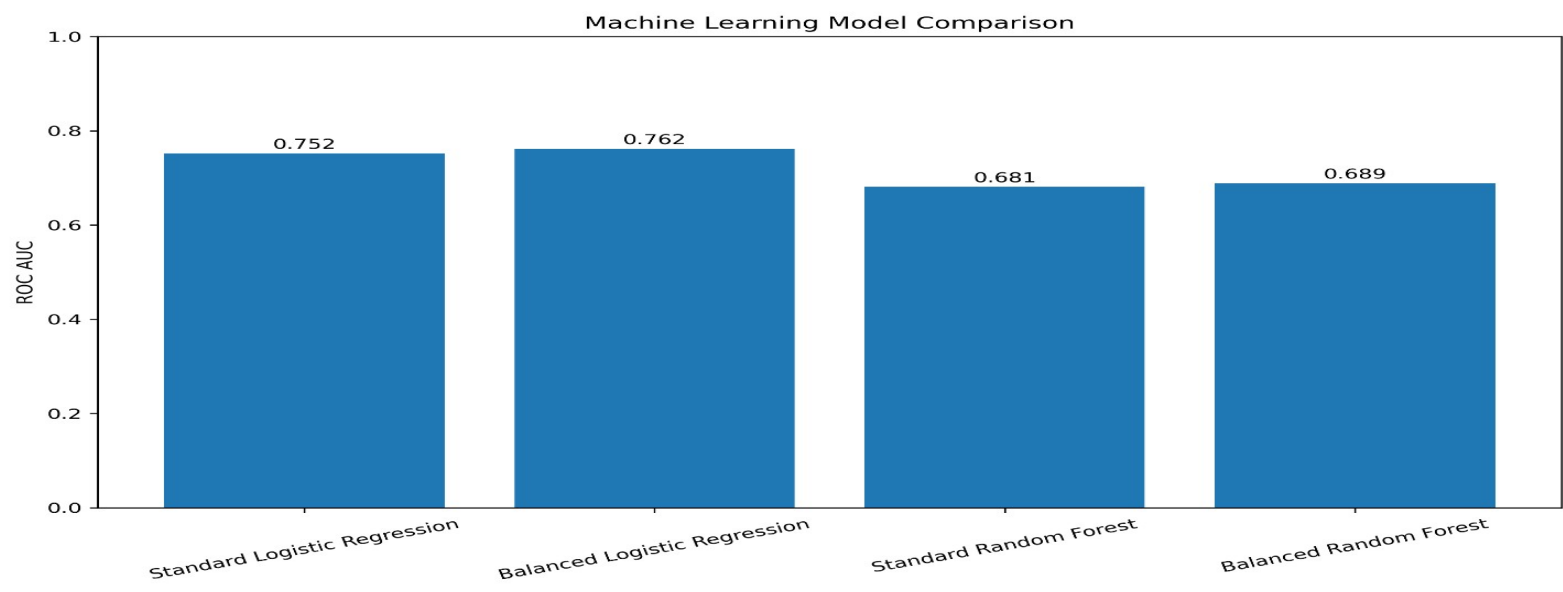
Months Since Last Delinquency

Key measure of stability & recovery.

Feature importance analysis revealed that credit risk is influenced by multiple financial and behavioural factors rather than any single borrower characteristic.

Model Comparison & Final Model Selection

Model	Accuracy	ROC-AUC	NP Detected	Selected
Linear Regression	—	—	—	✗
Standard Logistic Regression	98.2%	0.752	0 / 36	✗
Balanced Logistic Regression	69.7%	0.762	25 / 36	✓
Standard Random Forest	98.2%	0.681	0 / 36	✗
Balanced Random Forest	98.2%	0.689	0 / 36	✗



FINAL MODEL

Balanced Logistic Regression

- Highest ROC-AUC
- Detected 25 Defaults
- Most Practical
- Best Business Value

Interactive Power BI Dashboard

Objective

An interactive BI dashboard enabling stakeholders to monitor portfolio performance, identify credit risk patterns, and support data-driven lending decisions across six report pages.

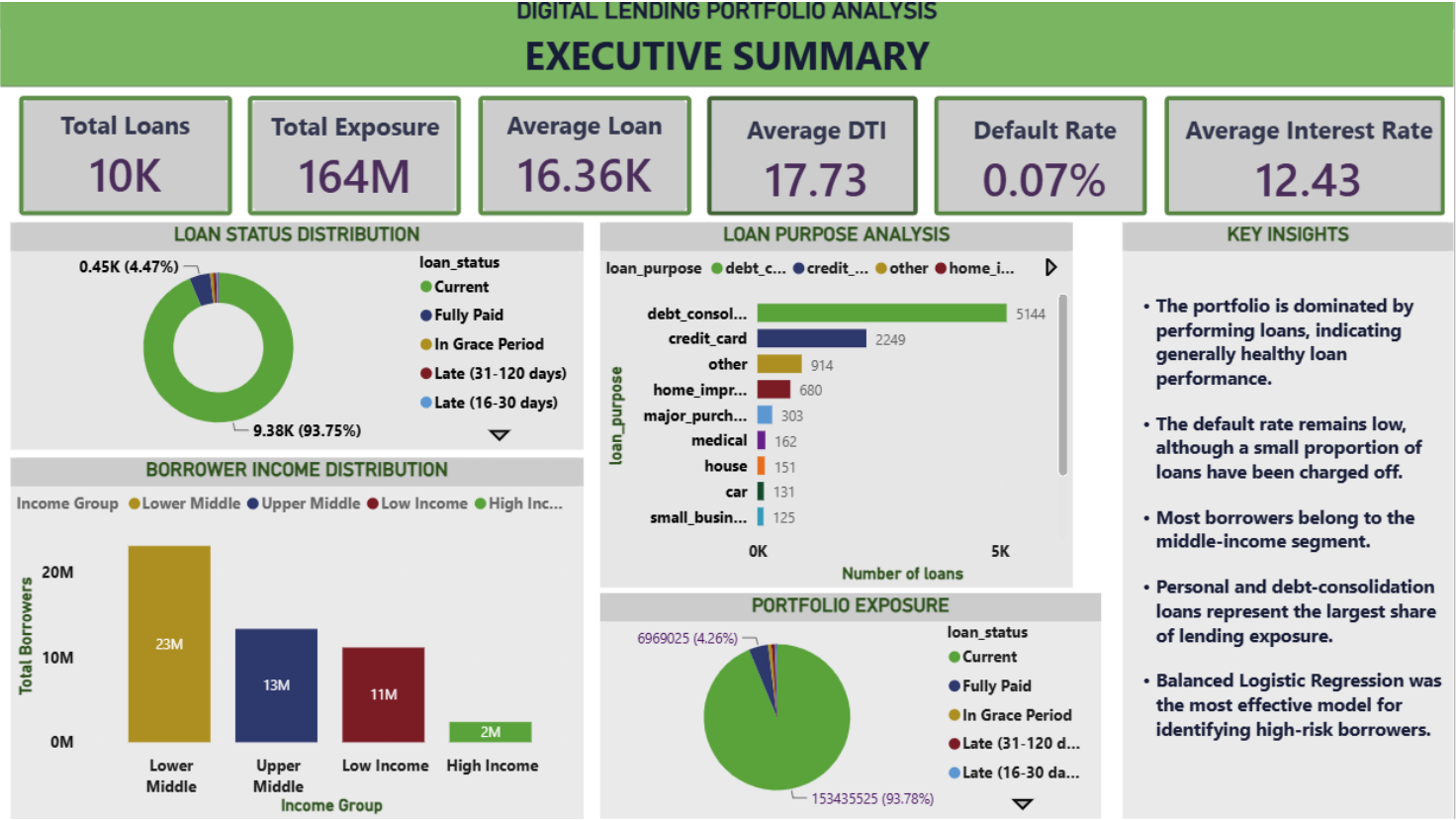
Dashboard Features

- Interactive filters and slicers
- Dynamic KPI cards
- Drill-down analysis
- Cross-filtering between visuals
- Executive-level reporting
- Machine learning integration



The Power BI dashboard transforms complex analytical results into intuitive, interactive visualizations that support faster and more informed lending decisions.

Executive Summary Dashboard



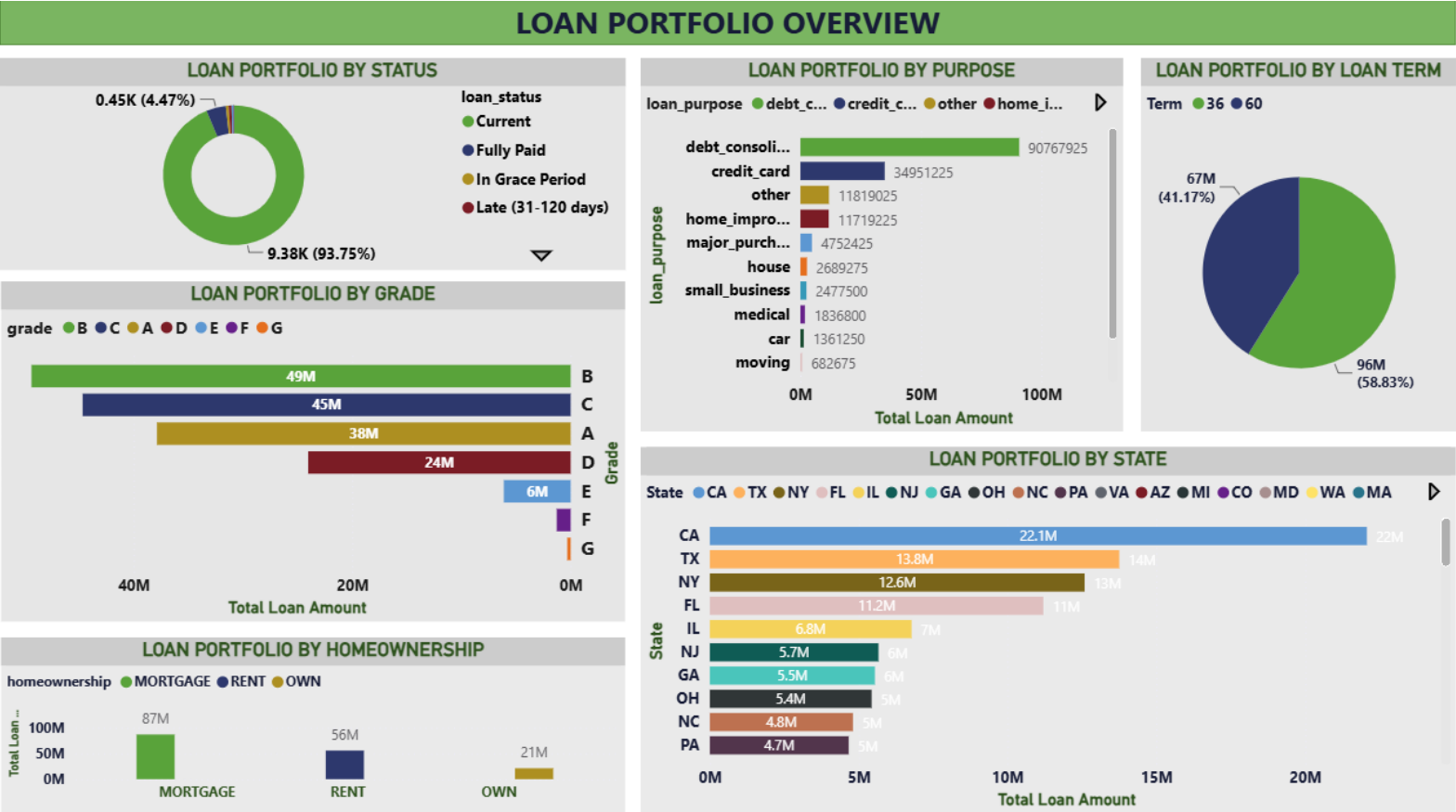
KPI	Value
Total Loans	10,000
Total Portfolio Exposure	\$164 Million
Performing Loans	9,822 (98.22%)
Non-Performing Loans	178 (1.78%)
Avg. Interest Rate	12.43%
Avg. Annual Income	\$89,138

Key Insights

- 98.22% of loans performing — strong portfolio quality
- \$164M in outstanding exposure requires ongoing monitoring
- 12.43% average rate reflects balanced, risk-based pricing
- Non-performing loans remain financially significant at 1.78%

The portfolio demonstrates strong overall performance with \$164 million in loan exposure and a 98.22% performing loan rate, while highlighting the need to continuously monitor non-performing loans.

Loan Portfolio Overview



LOAN PORTFOLIO BY HOMEOWNERSHIP

homeownership

MORTGAGE

RENT

OWN

100M

50M

0M

87M

56M

21M

MORTGAGE

RENT

OWN

LOAN PORTFOLIO BY STATE

State

CA

TX

NY

FL

IL

NJ

GA

OH

NC

PA

VA

AZ

MI

CO

MD

WA

MA

22.1M

13.8M

12.6M

11.2M

6.8M

5.7M

5.5M

5.4M

4.8M

4.7M

22M

14M

13M

11M

7M

6M

6M

5M

5M

5M

0M

5M

10M

15M

20M

Total Loan Amount

Current Loans

93.75% of Portfolio

Largest Loan Purpose

Debt Consolidation (\$90.8M)

Highest Grade Exposure

Grade B (\$49M)

Most Common Term

36 Months (58.83%)

Largest Homeownership

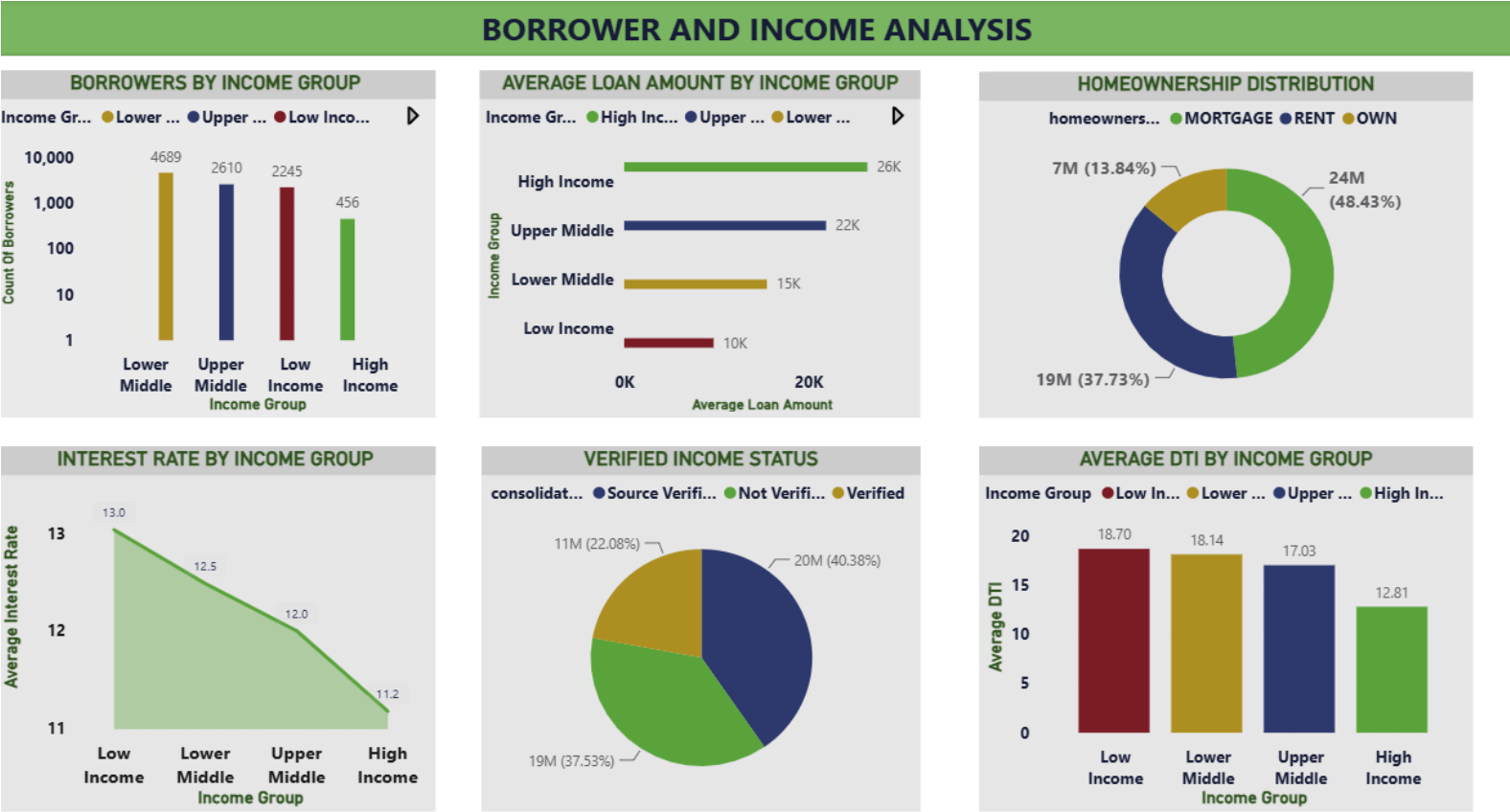
Mortgage Borrowers (\$87M)

Highest Geo Exposure

California (\$22.1M)

The loan portfolio is well-diversified across loan purposes, borrower profiles, credit grades, loan terms, and geographic regions, contributing to overall portfolio stability.

Borrower & Income Analysis



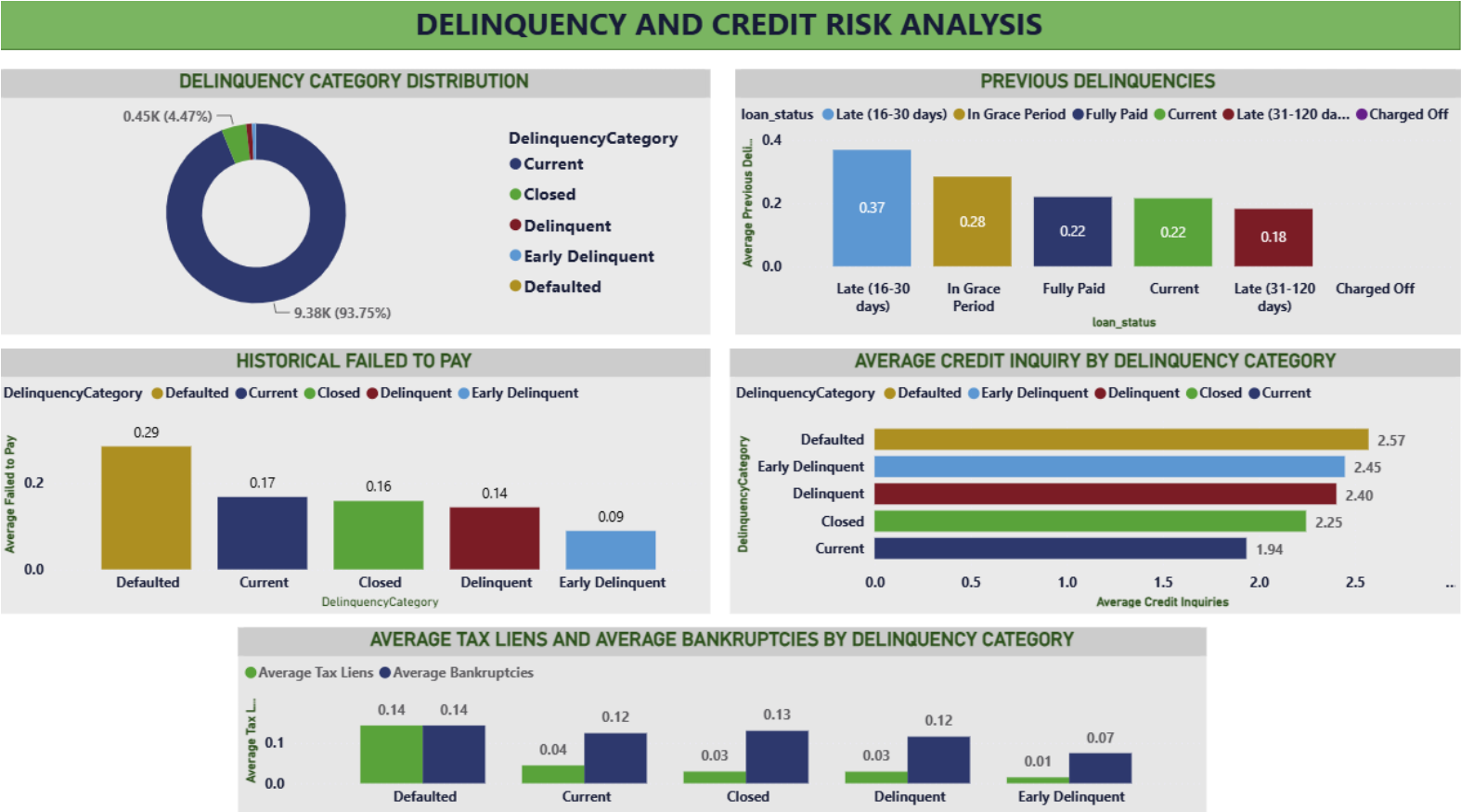
Income Group	Avg Loan	Avg Rate	Avg DTI
Low Income	\$10K	13.0%	18.70
Lower Middle	\$15K	12.5%	18.14
Upper Middle	\$22K	12.0%	17.03
High Income	\$26K	11.2%	12.81

Key Findings

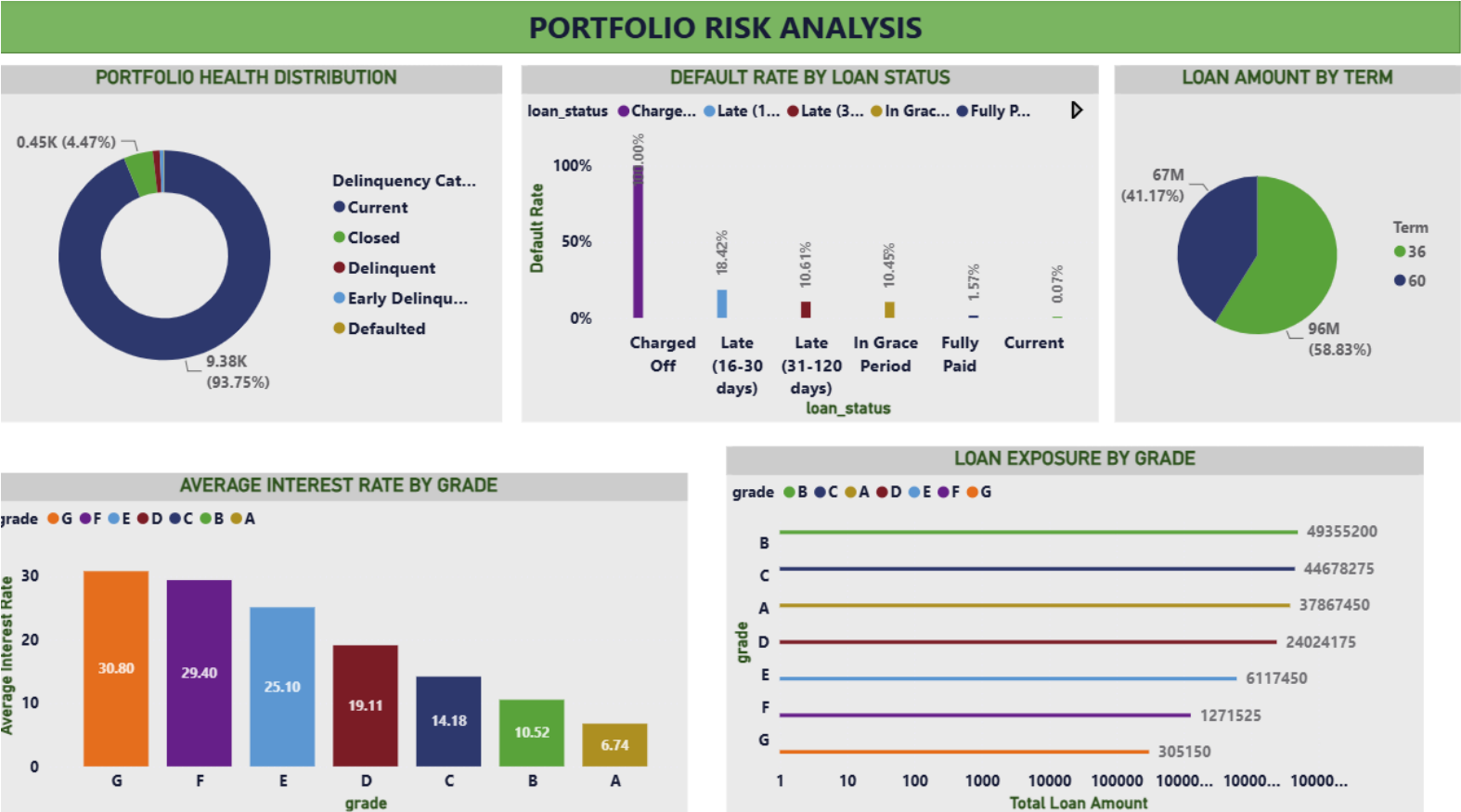
- Loan amount increases with income
- Interest rates decrease as income rises
- Higher-income borrowers carry lower DTI
- Mortgage is dominant homeownership status

Borrower income significantly influences loan size, interest rates, and debt burden, highlighting the importance of multi-factor creditworthiness assessment.

Delinquency & Credit Risk Analysis



Portfolio Risk Analysis



Current Loans

93.75% of Portfolio

Highest Default Rate

Charged Off (100%)

Largest Loan Exposure

Grade B (\$49.36M)

Most Common Term

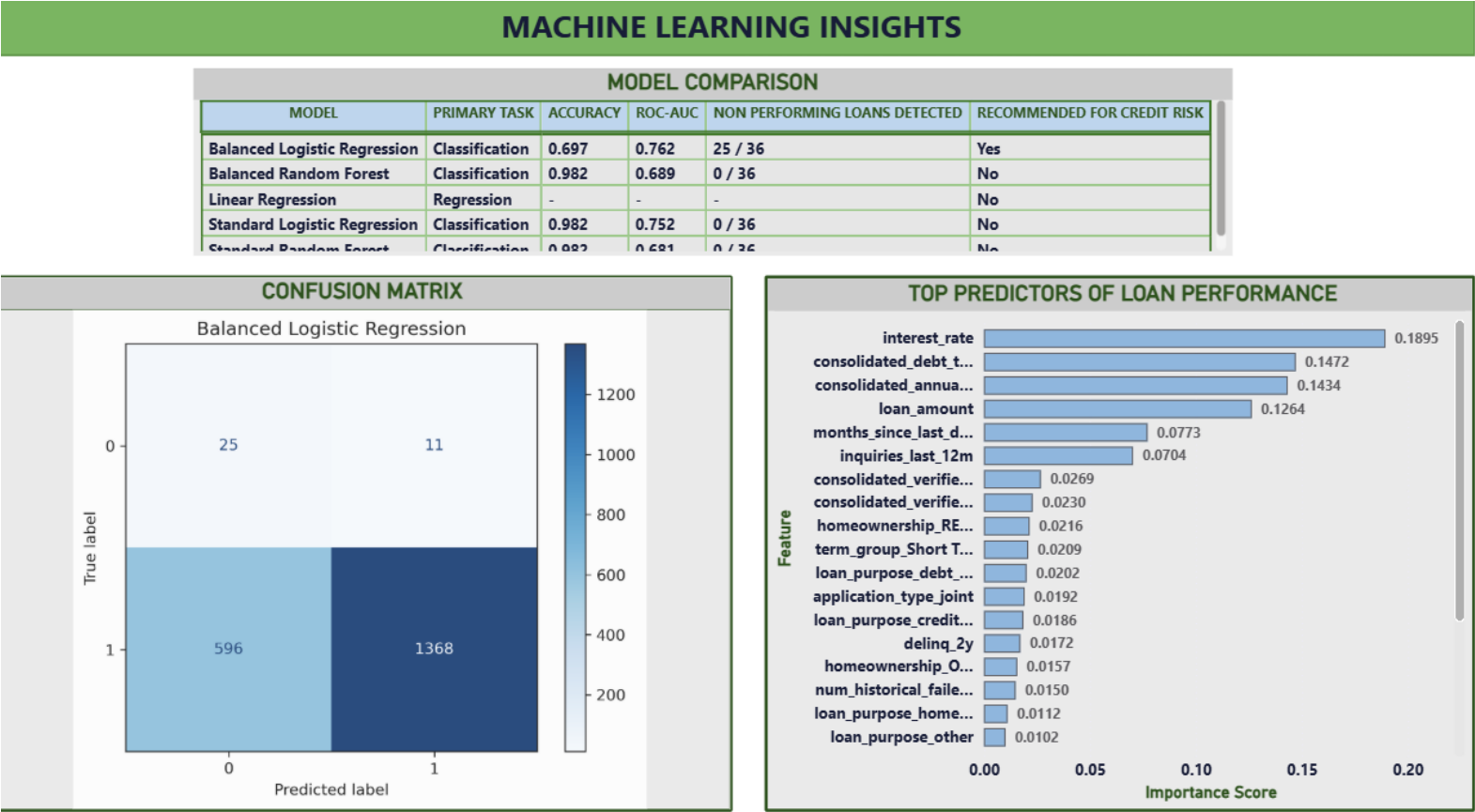
36 Months (58.83%)

Highest Interest Rate

Grade G (30.80%)

The portfolio demonstrates strong overall health, with lending concentrated in lower-risk credit grades and supported by an effective risk-based pricing strategy.

Machine Learning Insights



Best Model

Balanced Logistic Regression

ROC-AUC

0.762

High-Risk Loans Detected

25 of 36

Most Important Predictor

Interest Rate

Top Predictors

- Interest Rate
- Debt-to-Income Ratio
- Annual Income
- Loan Amount

Balanced Logistic Regression proved most effective because it prioritized identifying high-risk borrowers over overall accuracy, making it more suitable for real-world lending applications.

Business Recommendations



Adopt the Balanced Logistic Regression Model

Integrate it into the credit evaluation process — it best identifies high-risk borrowers.



Strengthen Borrower Risk Assessment

Use multiple characteristics: DTI, interest rate, income, delinquencies, inquiries.



Monitor Early Warning Indicators

Watch rising credit inquiries, delinquency history, high DTI, tax liens, bankruptcies.



Maintain a Balanced Lending Portfolio

Concentrate on Grades A-C; apply stricter underwriting to Grades D-G.



Continue Risk-Based Pricing

Higher-risk borrowers pay higher rates — ensures fair compensation and stability.

Expected Benefits

- Lower Default Risk
- Better Loan Approval Decisions
- Improved Portfolio Quality
- Higher Profitability
- Stronger Credit Risk Management

Combining descriptive analytics, interactive dashboards, and machine learning enables financial institutions to make more informed lending decisions and build a healthier, more resilient loan portfolio.

Conclusion & Future Work

Project Summary

This project combined data analytics, business intelligence, and machine learning to evaluate loan performance and identify the key factors influencing credit risk — transforming raw lending data into actionable insights.

Objectives Achieved

- Comprehensive data cleaning and preparation
- Exploratory data analysis of borrower and portfolio characteristics
- Interactive Power BI dashboards for portfolio monitoring
- Multiple ML models developed and compared for loan performance prediction
- Identified the most influential predictors of loan performance

Future Improvements

- Collect more recent lending data
- Apply advanced models (XGBoost, LightGBM, CatBoost)
- Explore probability-based risk scoring; incorporate macroeconomic indicators

Final Takeaways

- Strong Portfolio Health
- Data-Driven Decision Making
- Improved Credit Risk Identification
- Interactive Business Intelligence
- Predictive Machine Learning
- Actionable Business Recommendations

“Effective credit risk management is not about predicting every default — it is about identifying the right borrowers at the right time using data-driven insights.”

Thank You

Questions & Discussion

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